

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMC506 - APPLIED THERMODYNAMICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is a gaseous fuel? A) Diesel B) Petro C) Methane D) Kerosene
- 2 Which fuel has the highest calorific value? A) Coal B) Diesel C) Hydrogen D) Wood
- 3 Which of the following is a Primary fuel? A) Kerosene B) Petrol C) Wood D) LPG
- 4 Steam pressure is measured using a: A) Thermometer B) Venturimeter C) Bourdon gauge D) Orifice meter
- 5 Which boiler has high pressure and is used in modern power plants? A) Cochran boiler B) Lamont boiler C) Cornish boiler D) Simple vertical boiler
- 6 The Carnot cycle is a: A) Reversible cycle B) Irreversible cycle C) Semi-closed cycle D) Polytropic Cycle
- 7 The Otto cycle is used in: A) Steam power plant B) Gas Turbines C) Petrol engines D) Diesel engines
- 8 In Brayton cycle, the working fluid is : A) Steam B) Water C) Gas D) Refrigerant
- 9 The device used to remove air from the condenser is : A) Steam trap B) Air ejector C) Feed pump D) Injector
- 10 De Laval turbine is a type of: A) Reaction turbine B) Impulse turbine C) Mixed turbine D) Hydraulic turbine
- 11 The work input to a reciprocating compressor is minimum when the compression process is: A) Adiabatic B) Isothermal C) Isobaric D) Polytropic
- 12 In a vane type rotary compressor, vanes are: A) Stationary B) Sliding radially in slots C) Fixed to the shaft D) Reciprocating inside cylinder

Q2. Solve the following.

- A) Develop the equation for the measurement of calorific value of solid fuel by the use of bomb calorimeter with neat sketch. **6**
- B) Calculate the net and gross calorific value of a coal sample having following composition. C = 82%, H = 8%, O = 5%, N = 1.4% and ash = 3.6% **6**

Q3. Solve the following.

- A) Write about requirement of a good boilers. 6
- B) Dry saturated steam at 5 bar with negligible velocity expands in convergent nozzle to 1 bar and dryness fraction 0.94. Determine the velocity of steam leaving the nozzle 6
- Q4. Solve Any Two of the following.
- A) Steam expands from 2.5 bar to 1 bar in a nozzle. The initial velocity of steam is 80 m/s and initial temperature is 200°C. Taking nozzle efficiency as 90%, find the exit velocity. 6
- B) Derive an equation of Stirling cycle with neat sketch. 6
- C) Explain about super saturated flow through nozzles. 6
- Q5. Solve Any Two of the following.
- A) Write advantages and classification of steam turbines. 6
- B) List the types of cooling tower and explain any two of them with neat sketch. 6
- C) In a single stage impulse turbine, the blade angles are equal and the nozzle angle is 20°. The velocity coefficient for the blade is 0.83. Find the maximum blade efficiency possible. If the actual blade efficiency is 90% of maximum blade efficiency, find the possible ratio of blade speed to steam speed. 6
- Q6. Solve Any Two of the following.
- A) Explain in detail of single stage reciprocating air compressor with neat sketch. 6
- B) Write construction and working of root-blower. 6
- C) Explain the benefits of the multi staging in reciprocating air compressor with PV diagram. 6

*** End ***